Maharashtra State
Board of Technical Education
(ISO 9001:2008)

1983
S.S.V.P.S. ENGG & POLY. DHULE

Programme Name: Computer Engineering **Academic Year: 2026-27**

Programme Code: CO5K

Course Name and Code: Seminar and Project Initiation (315003)

A Seminar Report on

Title : Malicious URL Detection System for protecting user from Phishing Attacks

Submitted in partial fulfilment for the award of
THIRD YEAR DIPLOMA IN
COMPUTER ENGINEERING
By

Sr. No	Enrollment Number	Seat Number	Name of student
1	24610960192	388510	Ujwal Sandeep Chitte
2	24610960230	388538	Abhijit Sunil Patil
3	24610960237	388545	Janhavi Krishna Patil
4	24610960243	388550	Kimaya Nitin Patil

FOR ACADEMIC YEAR : 2026– 2027

Department of Computer Engineering

S.S.V.P.S's Bapusaheb Shivahirao Deore Polytechnic , Dhule

(Institute Code : 0059)

S.S.V.P.S's Bapusaheb Shivahirao Deore Polytechnic , Dhule

(An Affiliated Institute of Maharashtra State Board of Technical Education)

DEPARTMET OF COMPUTER ENGINEERING

Certificate

This is to certify that,

Roll No	Seat No	Full Name of Students	Enrollment No
23	388510	Ujwal Sandeep Chitte	24610960192
51	388538	Abhijit Sunil Patil	24610960230
58	388545	Janhavi Krishna Patil	24610960237
63	388550	Kimaya Nitin Patil	24610960243

of Fifth Semester Diploma in Computer Engineering of Institute Shri. Shivaji Vidya Prasarak Sanstha's Bapusaheb Shivajirao Deore Polytechnic,Dhule (Code:0059) has completed the Micro Project satisfactorily in Subject Seminar and Project Initiation (315003)for the academic year 2026 - 27 as prescribed in the MSBTE curriculum of K Scheme.

Place : Dhule

Date : .....

Signature:

Name : Shraddha S hajari

Subject Teacher

Signature:

Name : Nishad D Patel

Head of the Department

Seal of Institution

TABLE OF CONTENT

Sr. No.	Chapter	Page No.
	Abstract	i
	List of Figures	ii
	List of Tables	iii
1	Introduction	1
2	Literature Review	2
3	Information/Chapters	8
4	Advantages and Disadvantages	11
5	Conclusion	12
6	Project Initiation:	13
	6.1) Description of problem statement.	13
	6.2) Scope and objectives.	13
	6.3) Stake holder	13
	6.4) Platform/ Equipment/ Resources identification.	14
7	Seminar Presentation	15
	Bibliography	27
	References	28

ABSTRACT

The rapid growth of internet usage has increased the risk of cyber threats, particularly phishing attacks, fraudulent websites, and malware distribution through malicious URLs. Such URLs can mislead users into accessing unsafe websites and may result in the theft of sensitive information, including login credentials, banking details, and personal data. Traditional blacklist-based detection techniques have limitations in identifying newly generated or previously unknown malicious URLs. Therefore, an effective approach is required to identify potentially harmful URLs and improve online security.

The proposed project, **"**Malicious URL Detection Using Machine Learning," aims to classify URLs as legitimate or malicious using machine learning techniques. The system analyses relevant URL characteristics, such as URL length, the presence of special characters, domain-related information, and suspicious URL patterns. These features are extracted and processed to train and evaluate suitable machine learning models for URL classification. A web-based interface is proposed to allow users to enter a URL and obtain the predicted classification. The performance of the selected models will be assessed using evaluation metrics such as accuracy, precision, recall, and F1-score.

The proposed system seeks to provide a practical and accessible method for identifying potentially malicious URLs. By applying machine learning to URL analysis, the project aims to support the early detection of suspicious links, reduce exposure to phishing threats, and contribute to safer internet usage.

.

LIST OF FIGURES

Fig. No.	Title	Page No.
1.1	Flow of Malicious URL Detection System	10
1.2	Working of Malicious URL Detection System	10

LIST OF TABLES

Table No.	Title	Page No.
2.1	Literature Review	5
6.1	Resources	14

CHAPTER 1

INTRODUCTION

The internet has become part of daily life for banking, shopping, education, and communication. The Uniform Resource Locator (URL), the address used to reach any web resource, is the main way users move between websites, and attackers have learned to exploit it. A malicious URL is a link that leads to harmful content such as a fake login page, a site that silently downloads malware, a fraudulent redirect network, or a defaced website.

In India, the rapid growth of UPI payments and online banking has made these attacks more damaging. Attackers send links that imitate banks, delivery companies, government portals, and payment apps through WhatsApp, SMS, and email. Because the message often arrives from a known contact or looks urgent, many users click without checking. A single click can lead to stolen credentials, leaked OTPs, financial loss, or a malware infection.

Traditional protection depends on blacklists, which are lists of already-reported dangerous sites. These are reactive, because a new malicious domain can stay active for hours or days before it is reported. Machine learning offers a more proactive approach: a model can learn the patterns that separate malicious URLs from legitimate ones and judge a link it has never seen before.

This project proposes a Malicious URL Detection System that lets a user paste a link and get a quick verdict (safe, suspicious, or malicious). It combines a machine learning classifier with a cross-check against public threat-intelligence sources. The topic relates to Unit I (AI and ML in Digital Security) and Unit V (Digital Forensics and Ethical Hacking, including AI-powered phishing scams) of the Emerging Trends in Computer Engineering and Information Technology course.

CHAPTER 2

LITERATURE REVIEW

Research on malicious URL detection has evolved through three major approaches: blacklist-based and heuristic methods, machine-learning-based URL analysis, and advanced techniques using richer features and newer models.

2.1 Blacklist Based and Heuristic Methods

Early malicious URL detection systems mainly relied on manually maintained blacklists and predefined rules. A blacklist contains URLs that have already been identified as malicious. Heuristic methods use rules such as URL length, unusual characters, and the presence of an IP address instead of a domain name to identify suspicious links.

These methods are simple to implement and can quickly identify known malicious URLs. However, they may fail to detect newly created malicious websites that have not yet been reported or added to a blacklist. Attackers can also modify URL patterns to bypass predefined rules. Kytidou et al. (2025) discuss the challenges faced by traditional and modern phishing detection methods in keeping up with changing attack techniques.

2.2 Machine Learning-Based URL Detection

To overcome some limitations of traditional methods, researchers introduced supervised machine learning techniques. These techniques train classification models using datasets containing legitimate and malicious URLs.

URL detection features can be divided into three main categories:

1. **URL-based features:** URL length, special characters, domain structure, and suspicious character patterns.
2. **Domain-based features:** Domain age, IP address information, and other available domain characteristics.
3. **Webpage-based features:** HTML elements, JavaScript behaviour, and other webpage characteristics.

Research surveys, including work presented at ICT4SD 2025, discuss the use of these features for phishing URL detection. Machine learning algorithms such as Random Forest and Long Short-Term Memory (LSTM) networks have been explored for identifying malicious URL patterns. However, detection performance depends on the quality of the dataset, the selected features, and the evaluation method.

A study titled *Real-Time Phishing URL Detection Using Machine Learning* explores the use of a Random Forest classifier and the PhiUSIIL dataset available through the UCI Machine Learning Repository. This approach demonstrates the relevance of machine learning for developing URL classification systems intended to provide timely results.

2.3 Advanced Features Analysis and Modern Detection Models

Because attackers can modify URL strings to avoid detection, later research has explored additional information and more advanced learning methods.

Guo et al. (2025), in their work on graph-based phishing URL detection, investigate combining URL structure with network-level information, including IP addresses and authoritative name servers. Graph-based analysis considers relationships between related entities, potentially providing useful information beyond the URL string alone.

Similarly, *URL2Graph++: Unified Semantic-Structural-Character Learning for Malicious URL Detection* (2025) explores the combination of semantic, structural, and character-level features. This approach focuses on general malicious URL detection rather than phishing alone. Combining different feature types may help a model recognise a wider range of suspicious patterns.

However, advanced methods may require additional data, computational resources, and more complex processing. Their effectiveness also depends on the quality and availability of the information used.

2.4 Research Gap and Project Direction

The literature indicates that no single detection technique addresses every limitation of malicious URL detection. Blacklist-based methods may not recognise newly reported threats, while machine-learning models based on URL features can be affected by obfuscation and

changes in attack patterns. Graph-based and webpage-content-based approaches may require additional data and computational resources.

Another important consideration is accessibility. Ordinary users may need a simple tool that allows them to check a suspicious URL without understanding complex cybersecurity techniques.

Therefore, this project proposes studying a lightweight machine-learning classifier combined with a threat-intelligence cross-check through an easy-to-use web interface. The classifier would analyse URL characteristics, while the threat-intelligence component would check available information about known malicious URLs. The interface would present the result in a clear and understandable manner.

The proposed approach aims to make malicious URL checking more accessible to non-technical users. Its detection accuracy, efficiency, and practical usefulness would need to be evaluated through appropriate testing.

2.5 Conclusion of Literature Review

The reviewed research shows the development of malicious URL detection from traditional blacklists and heuristic rules to machine learning, graph-based analysis, and combined feature-learning approaches. Each technique offers different benefits and has its own limitations.

The literature supports further study of a practical detection approach that combines URL feature analysis with threat-intelligence information. This forms the foundation of the proposed Malicious URL Detection System for Protecting Users from Phishing Attacks.

Table No. 2.1: Literature Review:

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
1	Review-based evaluation of machine-learning techniques for malicious URL detection.	Machine-learning algorithms, URL feature extraction and classification techniques discussed across the reviewed studies.	91 research studies reviewed; individual datasets vary across the studies.	Review existing malicious URL detection methods, compare techniques and identify research challenges and future directions.	Different datasets, features and evaluation metrics make direct comparisons difficult. Results depend on the quality and coverage of the reviewed studies.
2	Machine-learning and quantum machine-learning approaches to malicious URL detection.	Machine learning, quantum machine learning and URL analysis techniques.	Datasets used in the paper's experiments; exact dataset names should be verified from the full paper.	Investigate methods for detecting malicious URLs, including phishing-related threats in an IoT security context.	Practical deployment and scalability of quantum approaches may be challenging. Results depend on dataset quality and experimental settings.
3	Deep learning using learned URL representations.	URLNet; character-level and word-level convolutional neural networks (CNNs).	Labeled malicious and legitimate URL datasets used by the original URLNet study; verify exact dataset names in the paper.	Learn URL representations automatically from characters and words to classify malicious and legitimate URLs.	Training may require substantial labeled data and computational resources. Performance on unseen URLs and evolving attack patterns requires further evaluation.
4	Machine-learning-based phishing detection using URL, domain and webpage-related features.	XGBoost, URL structure analysis, domain information and webpage behaviour analysis.	Specific dataset details are not provided in the available summary.	Detect phishing URLs by analysing URL structure, domain information and webpage behaviour.	Dataset details and testing procedures are insufficiently described to independently verify the reported 86.8% XGBoost accuracy.

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
5	Survey of malicious URL detection approaches.	URL analysis, blacklist-based detection and other detection techniques discussed in the survey.	Various studies and URL datasets discussed in the survey; no single dataset specified.	Identify malicious URLs, including phishing, spam and malware-related links, and address the limitations of blacklist-only methods.	The survey may not cover newer deep-learning techniques or recent phishing patterns; its coverage depends on its publication date.
6	Systematic literature review of phishing website detection techniques.	CNNs and other phishing website detection algorithms discussed in the reviewed studies.	PhishTank and datasets used in reviewed studies; 80 studies selected from 537 initial records.	Compare phishing website detection techniques, datasets, algorithms and reported performance.	Different datasets and evaluation methods make direct comparisons difficult. The reported 99.98% CNN accuracy is from one reviewed study and may not reflect real-world performance.
7	Phishing detection using machine learning (2020).	Principal Component Analysis (PCA), Support Vector Machine (SVM) and feature selection.	Exact dataset details are not provided in the supplied information.	Identify phishing URLs using feature selection and machine-learning classification.	Performance depends on selected features and dataset quality. Dataset details and evaluation results need verification against the original paper.
8	Malicious URL detection using distributed representations and deep learning (2022).	CNN, Bidirectional Gated Recurrent Unit (BiGRU), Bidirectional Long Short-Term Memory (BiLSTM), character-level and word-level representations	Exact dataset details are not provided in the supplied information.	Detect malicious URLs by learning character-level and word-level patterns with deep-learning models.	Deep-learning models may require substantial training data and computing resources. Generalisation to unseen URLs and emerging attacks requires evaluation.
9	Phishing detection from URLs using machine	Natural Language Processing (NLP),	Exact dataset details are not provided in	Detect phishing URLs by extracting and analysing URL-	Performance depends on feature quality and dataset diversity. Generalisation to newly created phishing

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
	learning (2019).	Random Forest and seven machine-learning classifiers.	the supplied information.	related features and comparing multiple classifiers.	URLs requires further testing.
10	Hybrid deep learning combining semantic, character-level and graph-based learning (URL2Graph++, 2025).	BERT, Character CNN, two Graph Convolutional Networks (GCNs) and a gated fusion network; compared with Dephides, CGAN, URLBERT and LLaMA-2.	GramBeddings (800,000 URLs), Mendeley (1,561,934 URLs) and Kaggle (632,503 URLs for cross-dataset testing).	Detect malicious URLs involving obfuscation, homoglyphs and shortened links, while improving generalisation across data sources.	Adaptive adversarial attacks are not fully tested. Webpage- and domain-level use is unvalidated. Training and inference costs may be high at large scale.
11	Graph-based machine learning using Loopy Belief Propagation (Guo et al., 2025).	Random Forest, Loopy Belief Propagation, Python 3.12, NetworkX, pandas, NumPy, scikit-learn, DNS queries, VirusTotal API, Word2Vec and Locally Linear Embedding.	One dataset from a prior study and one newly collected dataset; evaluated on 53,871, 100,000 and 306,354 samples. Dataset names were not supplied.	Improve detection robustness by modelling relationships among URLs, domains, IP addresses and nameservers, and improve graph-method convergence.	DNS and VirusTotal lookups may add latency and external-service dependence. Dataset names are unspecified, limiting reproducibility. Reported F1 was lower on the 53,871-sample set (94.73%) than the 98.77% reported on the largest set.
12	Real-time phishing URL detection using supervised machine-learning classifier comparison (2025).	RapidMiner, SMOTE, K-Nearest Neighbours (KNN), Naive Bayes, Decision Tree, Random Tree and Random Forest.	PhiUSIIL Phishing URL dataset (UCI): 235,795 instances and 54 attributes; 100,945 phishing and 134,850 legitimate instances.	Evaluate machine-learning classifiers for phishing detection and identify the best-performing classifier for real-time use.	Only one dataset is used, limiting evidence of generalisation. The reported 99.99% accuracy requires checks for data leakage and overfitting. Several features describe webpage properties, so the method is not strictly URL-only.

3

CHAPTER 3

INFORMATION

1 .Structure of the URL

A URL consists of protocol, subdomain, domain name, top-level domain, path, and optional query parameters. Attackers may manipulate these parts to create misleading links. For example, `paytm.secure-login.xyz` uses `secure-login.xyz` as its domain.

2. Types of Malicious URL

1. **Phishing URLs:** Steal passwords and personal information.
2. **Malware URLs:** Distribute harmful software.
3. **Defacement URLs:** Lead to hacked websites.
4. **Spam URLs:** Lead to unwanted advertisements or fraudulent redirects.

3. Common Evasion Techniques used by Attackers

Attackers use IP addresses, misspelled domains, shortened links, long URLs, newly registered domains, and misleading HTTPS websites to hide malicious links and avoid detection.

4. Feature Extraction

Feature extraction identifies URL characteristics for detection.

1. **Lexical features:** URL length, special characters, digits, and suspicious keywords.
2. **Host-based features:** Domain age, registration details, and DNS records.
3. **Content-based features:** Login forms, external links, and redirects.

5. Machine Learning Classification

Machine learning classifies URLs as legitimate or malicious. The process includes collecting labelled data, preprocessing URLs, extracting features, training models, and evaluating results. Common algorithms include Random Forest, Logistic Regression, Decision Tree, and SVM.

6. Datasets and Threat-Intelligence Sources

1. PhiUSIIL Phishing URL Dataset (UCI Repository)
2. Kaggle phishing and malicious URL datasets
3. PhishTank
4. Google Safe Browsing

These sources support model training and checking known malicious URLs.

7. Proposed System Architecture

The user submits a URL through a web interface. The system preprocesses the URL, extracts features, applies a machine-learning classifier, and checks available threat-intelligence sources. Finally, it displays the result as Safe, Suspicious, or Malicious, with a short explanation.

8. Performance Evaluation Metrices

1. **Accuracy:** Percentage of correct predictions.
2. **Precision:** How many URLs flagged as malicious are actually malicious.
3. **Recall:** How many actual malicious URLs are detected.
4. **F1-score:** Balance between precision and recall.

Recall is important for reducing missed threats, while precision helps reduce false alarms.

9. Tools and Technologies

The proposed system may use Python, Scikit-learn, Pandas, Flask or Streamlit, and Google Colab or Jupyter Notebook.

10. Legal and Ethical Aspects

The system should follow applicable cybersecurity and data-protection laws, including relevant provisions of the IT Act, 2000, and the DPDP Act, 2023. User-submitted URLs should be handled securely, stored only when necessary, and used for defensive cybersecurity purposes.

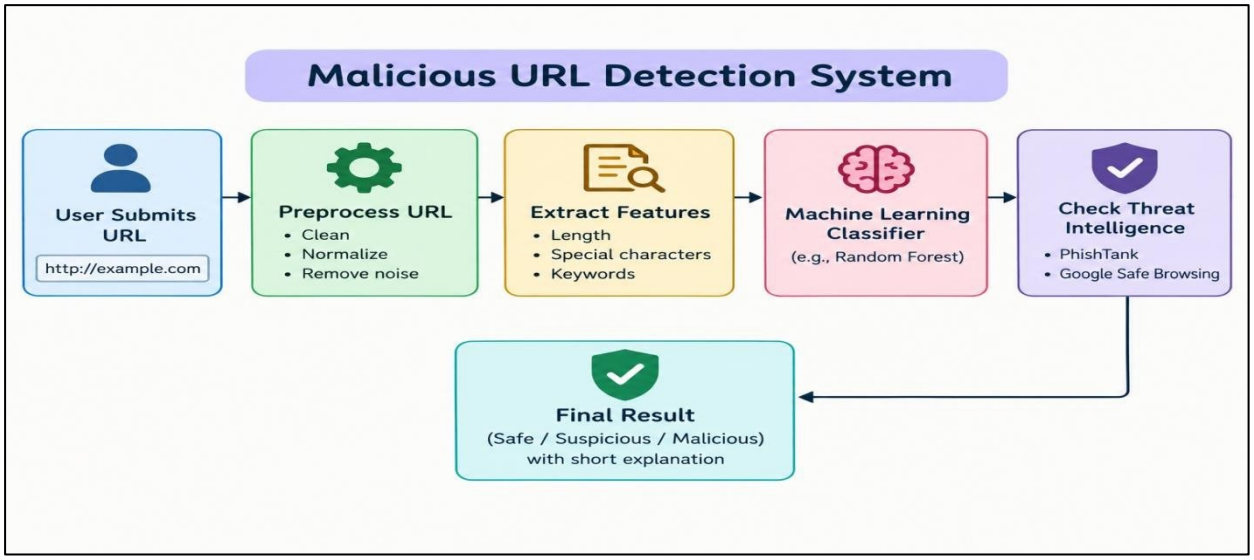


Fig 3.1:working of a malicious url detection system

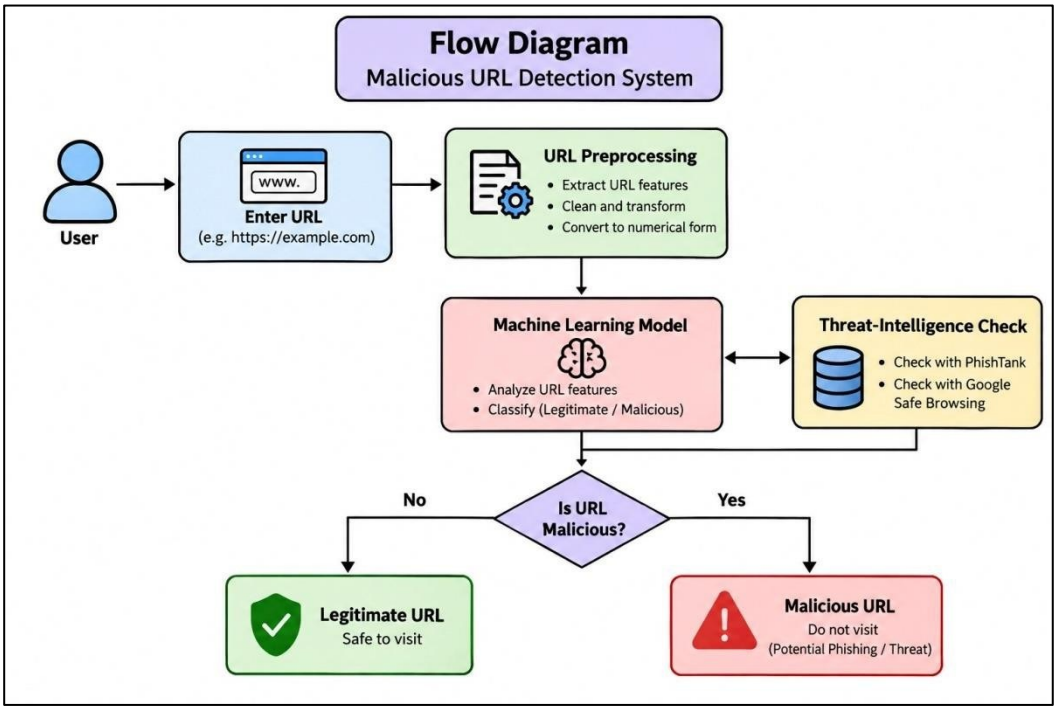


Fig 3.2:flow of a malicious url detection system

CHAPTER 4

ADVANTAGES AND DISADVANTAGES

4.1 Advantages

1. Detects potentially new malicious URLs using machine learning.
2. Provides quick results through a simple user interface.
3. Analyses URLs without directly opening suspicious webpages.
4. Uses open-source tools and datasets, reducing development costs.
5. Combines machine learning with blacklist checks for better detection.
6. Can be extended into a browser extension or messaging bot.
7. Improves awareness of phishing and online security.

4.2 Disadvantages

1. Accuracy depends on the quality and freshness of training data.
2. May incorrectly flag safe URLs or miss malicious ones.
3. Attackers can modify URLs to avoid detection.
4. Domain information checks may increase response time.
5. Cannot detect threats unrelated to URLs, such as malicious attachments or phone scams.
6. Newly created malicious websites may remain undetected.
7. Third-party services may have usage limits or change their policies.

CHAPTER 6
CONCLUSION

Malicious URLs, especially phishing links shared through emails, SMS, and messaging applications, are a major cybersecurity threat. They can trick users into revealing passwords, banking details, and other sensitive information. Traditional blacklist-based methods may fail to detect new malicious URLs that have not yet been reported.

This seminar studied machine-learning techniques for malicious URL detection using URL features and host-based information. The literature review explored different approaches, including Random Forest classification, graph-based detection, and threat-intelligence checks. Combining machine learning with threat intelligence may help improve the identification of suspicious URLs.

The proposed system aims to provide a simple and affordable way for users to check suspicious links before opening them. However, its performance depends on the quality of the training data, and incorrect predictions may occur.

In the future, the system could be extended with a browser extension, a messaging bot, QR-code scanning, and regular model updates. Overall, the proposed approach aims to improve phishing awareness and help users make safer decisions while browsing the internet.

CHAPTER 6
PROJECT INITIATION

6.1 Description of problem statement :

Malicious URLs are commonly used in phishing attacks to steal passwords, banking details, and other personal information. These links are shared through emails, SMS, and messaging applications. Traditional blacklist-based methods may fail to detect newly created malicious URLs because they have not yet been reported.

Therefore, there is a need for a system that analyses URL characteristics using machine learning and checks available threat-intelligence information to identify potentially malicious links and help users avoid online threats.

6.2 Scope and Objectives :

The project focuses on studying and proposing a malicious URL detection system using machine learning. It includes URL feature extraction, classification of legitimate and malicious URLs, and checking known threats using available threat-intelligence sources. The system aims to provide results through a simple web interface.

Objectives of the project:

1. To study different types of malicious URLs and phishing attacks.
2. To identify URL features useful for detecting suspicious links.
3. To study machine-learning algorithms for URL classification.
4. To combine URL analysis with threat-intelligence checks.
5. To provide understandable detection results through a web interface.
6. To improve user awareness of phishing and online security.

6.3 Stakeholders:

The main stakeholders of the proposed system are:

1. **Internet Users:** Check suspicious URLs before visiting them.
2. **Students and Researchers:** Study malicious URL detection techniques.
3. **Cybersecurity Professionals:** Use detection results as supporting security information.
4. **Educational Institutions:** Promote cybersecurity awareness.
5. **Organisations:** Explore the system for preliminary suspicious-link screening.
6. **Project Team:** Research, design, and evaluate the proposed system.

6.4 Platform / Equipment / Resources identification :

1. Equipment:

- **Computer or Laptop:** For developing and running the application.
- **Processor:** A standard modern processor suitable for Python development.
- **RAM:** Minimum 8 GB recommended; additional memory may be required for running local language models.
- **Internet Connection:** Required for accessing external AI services and downloading libraries.
- **Storage:** Sufficient space for source code, required libraries, and model files, if used.

2. Resoures:

Resource	Purpose
Computer or Laptop	Research and development
Internet Connection	Access research papers and datasets
Python	Machine-learning development
Pandas	Data processing and analysis
Scikit-learn	Training and evaluating models
Flask or Streamlit	Creating the web interface
Google Colab or Jupyter Notebook	Model training and experimentation
PhiUSIIL Dataset	Training and testing URL classification models
PhishTank / Google Safe Browsing	Checking known malicious URLs

CHAPTER 7

Seminar Presentation

	S.S.V.P.S's B. S. Deore Polytechnic, Dhule **Department of Computer Engineering** *Academic Year : 2026-2027*

SEMINAR AND PROJECT INITIATION COURSE

Sem : CO 5 K	**SPI – 315003**	**Date :**

Topic : Malicious URL Detection System to protect user from phishing attacks

Roll No.	Name of Student -
23	Ujwal Sandeep Chitte
51	Abhijit Sunil Patil
58	Janhavi Krishna Patil
63	Kimaya Nitin Patil

Contents

- Abstract

1. Introduction
2. Literature Review
3. Information to Seminar Topic
4. Advantages and Disadvantages
5. Conclusion
6. Project Initiation :
   - 6.1 Description of Problem Statement
   - 6.2 Scope and Objectives
   - 6.3 Stakeholder
   - 6.4 Platform/ Equipment/ Resources identification
7. Bibliography
8. References

Abstract

Malicious URLs, especially phishing links shared through WhatsApp, SMS, and email, are a common way for attackers to steal credentials and money, and most users cannot tell them apart from genuine links. Blacklist-based protection reacts only after a link is reported, so new malicious domains often go undetected. This seminar proposes a Malicious URL Detection System that uses a machine learning classifier (such as Random Forest), trained on URL features like length, special characters, IP usage, and domain age, along with a threat-intelligence cross-check. Users paste a link and get a Safe, Suspicious, or Malicious verdict through a simple web interface. The system is evaluated using accuracy, precision, recall, and F1-score.

1. Introduction

- The internet is used daily for banking, shopping, education, and communication, and the URL is the main way users reach any website.
- A malicious URL is a link that leads to harmful content, such as a fake login page, a malware download, a fraudulent redirect, or a defaced website.
- With the rapid growth of UPI payments and online banking in India, these attacks have become more damaging.
- Attackers send links that imitate banks, delivery companies, government portals, and payment apps through WhatsApp, SMS, and email.
- Messages often look urgent or arrive from a known contact, so many users click without checking.
- A single click can lead to stolen credentials, leaked OTPs, financial loss, or a malware infection.
- Traditional protection relies on blacklists of already-reported sites, which are reactive, so a new malicious domain can stay active for hours or days before it is blocked.

2. Literature Survey

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
1	Query-aware PII benchmark	LLMs; prompt strategies; benchmark tools	PII-Bench: 2,842 samples; 7 PII types; 55 categories	Test PII detection and query-based keep/mask decisions.	Relevance judgements weaker than detection; dense prompts remain difficult.
2	Privacy gateway with anonymization	Anonymization, substitution, verifiable processing	Not specified in supplied notes	Mask PII before third-party LLM use; restore values when needed.	Uniform masking ignores query relevance; context-aware filtering remains open.
3	Rule-based prompt sanitization	Rules, NER, local processing	Web-based LLM prompts; benchmark not specified	Sanitize sensitive prompt details before sending to online AI.	May hide details needed to answer the task.
4	Risk-adaptive browser prompt filtering	TypeScript, WebExtensions API, ONNX Runtime Web, regex, token classifier	Proposed precision, recall, F1, leakage and utility metrics	Score risk and allow, warn, redact or replace prompt content.	Browser coverage limited; results depend on detection and policy quality.
5	Client-side representation transformation	Python, transformer encoder, PyTorch, inference server	No fixed dataset; leakage, utility, bandwidth and latency tests proposed	Perturb prompt representations before compatible cloud inference.	May leak data or harm quality; requires server-side adaptation.
6	Prompt-threat detection with minimal logging	Python, FastAPI, text classifier, hashes, vector similarity	No fixed dataset; evaluation metrics proposed	Detect prompt injection and sensitive-data exposure with minimal logs.	Fingerprints may leak data or mismatch; detection alone is insufficient.

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
7	Enterprise API gateway and anomaly detection	OAuth 2.0, JWT, ML anomaly detection	Banking, healthcare, IoT; benchmark not specified	Secure API access and detect unauthorized activity or DDoS.	Adds latency; may produce errors; does not inspect PII in request content.
8	Federated learning with secure gateway	Kong Gateway, OAuth 2.0, JWT, TLS, differential privacy, zero-trust	CIFAR-10, MIMIC-III; 100 simulated edge nodes	Protect federated learning while keeping raw data local.	Privacy adds accuracy, communication and latency costs; scalability needs testing.
9	Sensitive-data replacement	Tokenization, placeholders, reversible mapping	Dataset details not fully visible in supplied image	Replace sensitive values before sending prompts to external models.	Encoding may be exposed or context may be lost if mapping is mishandled.

7

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
10	Context-aware PII detection	NER, contextual classifiers, policy rules	Not visible in supplied image	Identify which personal data is relevant to the user's request.	False positives and missed entities can reduce utility and privacy.
11	Prompt-injection and data-leakage screening	Input classifiers, pattern rules, security filters	Not visible in supplied image	Detect unsafe instructions and possible disclosure before model processing.	Evasion and new attack patterns can bypass static filters.
12	Privacy-preserving LLM pipeline	Local preprocessing, redaction, secure API, audit controls	Not visible in supplied image	Combine detection, redaction and controlled model access in one workflow.	Extra processing can increase latency; needs broad real-world evaluation.

8

3. Information to seminar topic

1. Structure of a URL

- A URL contains protocol, domain name, extension, path, and parameters.
- Attackers use fake or similar-looking domains to trick users.

2. Types of Malicious URLs

- Phishing URLs – Steal passwords and personal details.
- Malware URLs – Download harmful software.
- Defacement URLs – Show modified or hacked websites.
- Spam URLs – Lead to unwanted or fraudulent websites.

3. Common Evasion Techniques

- Fake or misspelled domain names.
- Shortened URLs and long, complex links.
- Newly registered domains and misleading HTTPS links.

9

3. Information to seminar topic

4. Feature Extraction

- Lexical Features: URL length, dots, numbers, symbols, and keywords.
- Host-based Features: Domain age, DNS records, and registration details.

5. Machine Learning Classification

- Collect and clean URL datasets.
- Extract features and train ML models.
- Use Random Forest, Decision Tree, or SVM.

6. Datasets and Tools

- Datasets: UCI, Kaggle, and PhishTank.
- Tools: Python, Pandas, Scikit-learn, and Flask/Streamlit.

10

3. Information to seminar topic

7. Proposed System Architecture

- URL Input → Pre-processing → Feature Extraction → ML Model → Blacklist Check → Final Result.
- Output: Safe, Suspicious, or Malicious.

8. Performance Evaluation

- Accuracy – Overall correct predictions.
- Precision – Correct malicious URL predictions.
- Recall – Malicious URLs detected.
- F1-score – Balance between precision and recall.

9. Legal and Ethical Aspects

- Follow applicable cybercrime and data protection laws.
- Protect user privacy and use the system only for defensive purposes.

11

3. Information to seminar topic

7. Proposed System Architecture

- URL Input → Pre-processing → Feature Extraction → ML Model → Blacklist Check → Final Result.
- Output: Safe, Suspicious, or Malicious.

8. Performance Evaluation

- Accuracy – Overall correct predictions.
- Precision – Correct malicious URL predictions.
- Recall – Malicious URLs detected.
- F1-score – Balance between precision and recall.

9. Legal and Ethical Aspects

- Follow applicable cybercrime and data protection laws.
- Protect user privacy and use the system only for defensive purposes.

11

4. Advantages / Disadvantages

Advantages

- **Detects new attack**s: It can detect new malicious URLs that are not in the blacklist.
- **Fast results:** It checks URLs within seconds and is easy to use.
- **Safe checking:** It checks the URL without opening the harmful website.
- **Low cost:** It uses free tools and public datasets.

Disadvantages

- **Dataset dependency:** Accuracy depends on the quality of the training data.
- **Wrong results:** It may mark a safe URL as harmful or miss a harmful URL.
- **Limited detection:** Some specially designed malicious URLs may escape detection.
- **Slow checking:** Checking domain information may take extra time.

12

5. Conclusion

- Malicious URLs and phishing links are used to steal personal information, passwords, and money. Traditional methods may not detect new threats. Machine Learning helps identify malicious URLs by checking their features and patterns.

- The proposed system allows users to check suspicious links quickly and easily before clicking them. Combining Machine Learning with a blacklist can improve detection. In the future, the system can be improved by adding browser extensions, WhatsApp or Telegram bots, QR-code scanning, and regular updates to detect new phishing attacks and improve online safety.

13

6. Project Initiation

- Identify the problem of phishing attacks through malicious URLs.
- Understand the need for detecting fake and suspicious websites.
- Develop a system to analyze URLs and detect potential threats.
- Classify URLs as legitimate or malicious.
- Warn users about potentially dangerous links.
- Aim to protect users' personal information and improve online safety.

14

6.1 Problem Statement

- Malicious URLs are used in phishing attacks to steal passwords, banking details, and personal information.
- These links are shared through emails, SMS, and messaging applications.
- Traditional blacklist methods may fail to detect newly created malicious URLs.
- Therefore, a system is needed to analyze URL features using machine learning and threat intelligence.
- The system aims to identify malicious URLs and warn users about potential phishing threats.

15

6.2 Scope and Objectives

Scope:

- Detect potentially malicious and legitimate URLs.
- Extract URL features and analyze suspicious links.
- Use machine learning and threat-intelligence sources.
- Display detection results through a simple web interface.

Objectives:

- To study phishing attacks and malicious URLs.
- To identify important URL features.
- To study machine-learning algorithms for URL classification.
- To check known threats using threat-intelligence sources.
- To display clear detection results through a web interface.
- To create awareness about phishing and online security.

16

6.3 Stake Holder

- **Internet Users:** Check suspicious URLs before opening them.
- **Students and Researchers:** Study malicious URL detection techniques.
- **Cybersecurity Professionals:** Use detection results for security support.
- **Educational Institutions:** Promote cybersecurity awareness.
- **Organisations:** Screen suspicious links for potential threats.
- **Project Team:** Design, develop, and evaluate the system.

17

6.4 Platform / Equipment / Resources identification

- **Computer or Laptop:** For research and development.
- **Internet Connection:** To access research papers and datasets.
- **Python:** For machine-learning development.
- **Pandas:** For data processing and analysis.
- **Scikit-learn:** For training and evaluating machine-learning models.
- **Flask or Streamlit:** For creating the web interface.
- **Google Colab or Jupyter Notebook:** For model training and experimentation.
- **PhiUSIIL Dataset:** For training and testing URL classification models.
- **PhishTank / Google Safe Browsing:** For checking known malicious URLs.

18

Bibliography

- Google for Developers, "Google Safe Browsing API." Available at: https://developers.google.com/safe-browsing/reference
- Scikit-learn, "Machine Learning in Python." Available at: https://scikit-learn.org/
- Pandas, "Python Data Analysis Library." Available at: https://pandas.pydata.org/
- Streamlit, "Build Data Apps." Available at: https://docs.streamlit.io/
- Flask, "Flask Web Development Framework." Available at: https://flask.palletsprojects.com/
- PhiUSIIL Phishing URL Dataset, "Phishing URL Classification Dataset." Available at: https://archive.ics.uci.edu/

19

References

- M. Aljabri et al., "Detecting Malicious URLs Using Machine Learning Techniques: Review and Research Directions," IEEE Access, vol. 10, pp. 121395–121417, 2022. https://doi.org/10.1109/ACCESS.2022.3222307
- H. Le, Q. Pham, D. Sahoo, and S. C. H. Hoi, "URLNet: Learning a URL Representation with Deep Learning for Malicious URL Detection," Proceedings of the 2018 ACM International Conference on Web Search and Data Mining, 2018. https://arxiv.org/abs/1802.03162
- N. Reyes-Dorta, P. Caballero-Gil, and C. Rosa-Remedios, "Detection of Malicious URLs Using Machine Learning," Wireless Networks, vol. 30, pp. 7543–7560, 2024. https://doi.org/10.1007/s11276-024-03700-w
- D. Sahoo, C. Liu, and S. C. H. Hoi, "Malicious URL Detection Using Machine Learning: A Survey," arXiv preprint arXiv:1701.07179, 2017. https://arxiv.org/abs/1701.07179
- F. Vanhoenshoven, G. Nápoles, R. Falcon, K. Vanhoof, and M. Köppen, "Detecting Malicious URLs Using Machine Learning Techniques," IEEE Symposium Series on Computational Intelligence, pp. 1–8, 2016.
- O. K. Sahingoz, E. Buber, O. Demir, and B. Diri, "Machine Learning Based Phishing Detection from URLs," Expert Systems with Applications, vol. 117, pp. 345–357, 2019. https://doi.org/10.1016/j.eswa.2018.09.029

20

References

- J. Ma, L. K. Saul, S. Savage, and G. M. Voelker, "Beyond Blacklists: Learning to Detect Malicious Web Sites from Suspicious URLs," Proceedings of the 15th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 1245–1254, 2009.
- S. Garera, N. Provos, M. Chew, and A. D. Rubin, "A Framework for Detection and Measurement of Phishing Attacks," Proceedings of the ACM Workshop on Recurring Malcode, pp. 1–8, 2007.
- A. Le, A. Markopoulou, and M. Faloutsos, "PhishDef: URL Names Say It All," Proceedings of IEEE INFOCOM, pp. 191–195, 2011.
- R. Verma and K. Dyer, "On the Character of Phishing URLs: Accurate and Robust Statistical Learning Classifiers," Proceedings of the 5th ACM Conference on Data and Application Security and Privacy, pp. 111–122, 2015.
- F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011. https://jmlr.org/papers/v12/pedregosa11a.html
- T. M. Mitchell, Machine Learning. McGraw-Hill, 1997.
- A. C. Müller and S. Guido, Introduction to Machine Learning with Python: A Guide for Data Scientists. O'Reilly Media, 2016.

21

Thank you

22

BIBLIOGRAPHY

❖ **Books**

1. Artificial Intelligence: A Modern Approach — Stuart Russell and Peter Norvig.
2. Machine Learning — Tom M. Mitchell.
3. Introduction to Machine Learning with Python — Andreas C. Müller and Sarah Guido.
4. Security Engineering — Ross Anderson.

❖ **Websites**

1. Google Safe Browsing — https://safebrowsing.google.com/
2. OpenPhish — https://openphish.com/
3. PhishTank — https://phishtank.org/
4. IEEE Xplore Digital Library — https://ieeexplore.ieee.org/
5. ACM Digital Library — https://dl.acm.org/
6. arXiv — https://arxiv.org/

REFERENCES

1. M. Aljabri et al., “Detecting Malicious URLs Using Machine Learning Techniques: Review and Research Directions,” IEEE Access, vol. 10, pp. 121395–121417, 2022. https://doi.org/10.1109/ACCESS.2022.3222307

2. H. Le, Q. Pham, D. Sahoo, and S. C. H. Hoi, “URLNet: Learning a URL Representation with Deep Learning for Malicious URL Detection,” Proceedings of the 2018 ACM International Conference on Web Search and Data Mining, 2018. https://arxiv.org/abs/1802.03162

3. N. Reyes-Dorta, P. Caballero-Gil, and C. Rosa-Remedios, “Detection of Malicious URLs Using Machine Learning,” Wireless Networks, vol. 30, pp. 7543–7560, 2024. https://doi.org/10.1007/s11276-024-03700-w

4. D. Sahoo, C. Liu, and S. C. H. Hoi, “Malicious URL Detection Using Machine Learning: A Survey,” arXiv preprint arXiv:1701.07179, 2017. https://arxiv.org/abs/1701.07179

5. F. Vanhoenshoven, G. Nápoles, R. Falcon, K. Vanhoof, and M. Köppen, “Detecting Malicious URLs Using Machine Learning Techniques,” IEEE Symposium Series on Computational Intelligence, pp. 1–8, 2016.

6. O. K. Sahingoz, E. Buber, O. Demir, and B. Diri, “Machine Learning Based Phishing Detection from URLs,” Expert Systems with Applications, vol. 117, pp. 345–357, 2019. https://doi.org/10.1016/j.eswa.2018.09.029

7. J. Ma, L. K. Saul, S. Savage, and G. M. Voelker, “Beyond Blacklists: Learning to Detect Malicious Web Sites from Suspicious URLs,” Proceedings of the 15th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 1245–1254, 2009.

8. S. Garera, N. Provos, M. Chew, and A. D. Rubin, “A Framework for Detection and Measurement of Phishing Attacks,” Proceedings of the ACM Workshop on Recurring Malcode, pp. 1–8, 2007.

9. A. Le, A. Markopoulou, and M. Faloutsos, “PhishDef: URL Names Say It All,” Proceedings of IEEE INFOCOM, pp. 191–195, 2011.

10. R. Verma and K. Dyer, “On the Character of Phishing URLs: Accurate and Robust Statistical Learning Classifiers,” Proceedings of the 5th ACM Conference on Data and Application Security and Privacy, pp. 111–122, 2015.
11. T. Peng, I. Harris, and Y. Sawa, “Detecting Phishing Attacks Using Natural Language Processing and Machine Learning,” Proceedings of the IEEE 12th International Conference on Semantic Computing, pp. 300–301, 2018.
12. F. Pedregosa et al., “Scikit-learn: Machine Learning in Python,” Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011. https://jmlr.org/papers/v12/pedregosa11a.html
13. T. M. Mitchell, Machine Learning. McGraw-Hill, 1997.
14. A. C. Müller and S. Guido, Introduction to Machine Learning with Python: A Guide for Data Scientists. O’Reilly Media, 2016.
15. C. M. Bishop, Pattern Recognition and Machine Learning. Springer, 2006.
16. S. Russell and P. Norvig, Artificial Intelligence: A Modern Approach, 4th ed. Pearson, 2021.
17. Python Software Foundation, “Python Documentation.” https://docs.python.org/3/
18. Scikit-learn Developers, “Scikit-learn User Guide.” https://scikit-learn.org/stable/user_guide.html
19. Google, “Google Safe Browsing.” https://safebrowsing.google.com/
20. OpenPhish, “OpenPhish: Phishing Intelligence.” https://openphish.com/
21. PhishTank, “PhishTank: An Anti-Phishing Site.” https://phishtank.org/
22. UCI Machine Learning Repository, “Machine Learning Datasets.” https://archive.ics.uci.edu/
23. IEEE, “IEEE Xplore Digital Library.” https://ieeexplore.ieee.org/
24. Association for Computing Machinery, “ACM Digital Library.” https://dl.acm.org/

SEMINAR and PROJECT INITIATION Assessment / Evaluation

Roll No	Seat No	Full Name of Students	Enrollment No
23	388510	Ujwal Sandeep Chitte	24610960192
51	388538	Abhijit Sunil Patil	24610960230
58	388545	Janhavi Krishna Patil	24610960237
63	388550	Kimaya Nitin Patil	24610960243

Formative Assessment

Sr. No.	Criteria	Weightage	Exam Seat Number			
			388510	388538	388545	388550
1	Selection Topic / Theme of seminar	5				
2	Literature review and data presentation	5				
3	Quality of Preparation and innovativeness	5				
4	Q-A handling	5				
5	Time Management	5				
6	Seminar Presentation report	10				
7	Selection of Theme of Problem Statement and its innovativeness	5				
8	Stages of development of Action plan	5				
9	Prototyping	5				
	Total	**50**				
	Scaled to 25					

Sign :

Name : ----------

(Course Expert/s)